

JAYRAJ M. PAMNANI

jmp10051@nyu.edu | [Linkedin](#) | [Github](#) | [About Me](#) | +1(646) 642-1043

Work Authorization: F1 OPT (12 Months) | STEM Extension Eligible (36 Months Total)

SUMMARY

Highly skilled Cloud, DevOps, and MLOps Engineer specializing in Azure, AWS, and GCP. Expertise includes Kubernetes, advanced CI/CD, and designing secure, production-grade and containerized infrastructure. Proven track record of improving reliability, reducing costs, and accelerating release cycles for both software and machine learning pipelines.

EDUCATION

New York University | Master's in Computer Engineering | May 2026

Relevant Coursework: High Performance Machine Learning, Machine Learning Operations, Deep Learning, Database Systems, Big Data

Parul University | Bachelor's in Computer Science & Engineering with Specialization in AI | May 2024

Relevant Coursework: Data Structures and Algorithms, Natural Language Processing, GPU computing, Pattern Recognition, Object Oriented Programming, Computer Networks

SKILLS

DevOps & MLDevelopment: Ansible, Grafana, Prometheus, Terraform, MLflow, TorchServe, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), REST APIs, Django, Flask, NextJS, Kafka, Airflow, WandB

AI/ML & Data Science: Python, R, TensorFlow, PyTorch, Scikit-learn, Keras, LangChain, LLMs, RAG, CUDA

Big Data & Analytics: Spark, BigQuery, PostgreSQL, MongoDB, Pandas, NumPy, Scala, PySpark, Tableau, PowerBI, Snowflake

Tools & Platforms: AWS (SageMaker), GCP (Vertex AI), Azure, HuggingFace, Ollama, GitHub, Jupyter, n8n, Postman

PROFESSIONAL EXPERIENCE

GBCS Group - Software Architect Intern, Calgary, Canada

11/2025 - 02/2026

- Drove cloud infrastructure cost-efficiency by implementing resource optimization strategies across the software ecosystem, resulting in **28% reduction in hosting and maintenance costs**.
- Reengineered automated CI/CD pipelines and deployment workflows, **accelerating release cycles by 40%** while ensuring 99.9% system availability.
- Defined technical standards and architectural roadmaps, guiding development teams in building scalable microservices that reduced technical debt and improved maintainability by 30%.

New York University - Teaching Assistant – Machine Learning, New York, NY

09/2025 – 12/2025

- Guided **50+** graduate students through ML fundamentals, including preprocessing pipelining, supervised/unsupervised learning, Deep Learning, and model optimization techniques.
- Conducted weekly office hours to debug Python code, explained algorithms, taught ML topics, and assisted with PyTorch implementations.

Swaroop.ai - AI Intern, Ahmedabad, India

03/2024 - 08/2024

- Enhanced Text-to-Speech model performance by **25%** through **fine-tuning Coqui TTS** across multiple Indian languages, optimizing real-time synthesis pipelines.
- Improved Speech-to-Text accuracy by **18%** by refining Coqui STT models with domain-specific audio datasets and custom preprocessing scripts.
- Integrated **OpenAI Whisper** into the production workflow, increasing transcription accuracy on noisy datasets by **30%** and cutting **inference latency by 20%**.
- Streamlined the **end-to-end MLOps pipeline** (data cleaning, model training, deployment orchestration) to enhance velocity, reducing model turnaround time from **3 days to under 24 hours**.

Freelance Software Developer, Vadodara, India

07/2023 – 09/2023

- Engineered and deployed a Django-based web app that automated data transfer between **QuickBooks and KatanaMRP**, cutting the client's manual bookkeeping workload by **~85%**.
- Built a robust **REST API integration** using Django (backend) and JavaScript (frontend), enabling seamless synchronization of **10,000+ financial and inventory records** with zero data loss.
- Designed and implemented **5+ custom verification layers**, achieving **99.8% data accuracy** before syncing to KatanaMRP.
- Deployed on **AWS EC2** with automated error logging and uptime above **99.9%**, ensuring secure, continuous operation for the client's business processes.

PROJECTS

Transaction Category Prediction | PyTorch, MLflow, QLoRA, Kubernetes, FastAPI, CI/CD

- Built an **end-to-end MLOps pipeline** for transaction category prediction, fine-tuning transformer NLP models with LoRA & QLoRA across multiple NVIDIA GPUs on Chameleon Cloud bare-metal IaaS.
- Implemented **CI/CD workflows** with model monitoring and drift detection, enabling **continuous re-training via Kubernetes orchestration** and FastAPI-served inference endpoints.

HexDrop (Secure File Transferring WebApp) | Next.js, TypeScript, Prisma, PostgreSQL, AWS, Docker, K8s

- Built a secure file-sharing app where users upload files, receive a shareable 6-digit key, and download via that key; implemented end-to-end flow with client-side encryption, Prisma-managed metadata in PostgreSQL, and object storage in AWS S3.
- Designed and deployed a full-stack **DevOps pipeline**: multi-stage Docker builds, **GitHub Actions CI/CD** for build and deploy, **EKS orchestration** with HPA and ALB Ingress, and RDS PostgreSQL with External Secrets for configuration management.

EyeConnect | WebRTC, Supabase, OpenRouter AI, React, TypeScript

- Developed an end-to-end accessibility platform that connects blind users with sighted volunteers using **WebRTC** for peer-to-peer video streaming and **Supabase Realtime** for instant signaling and volunteer matching; awarded **2nd place at NYU Hacks 2025**.
- Architected a hybrid assistance system integrating **OpenRouter AI** for automated vision descriptions and a **React/TypeScript** frontend with **shadcn/ui** to ensure a high-contrast, accessible UI for low-latency visual support.